

SpaceX Falcon 9 First-Stage Landing Analytics

An End-to-End Data Acquisition, Exploratory Analysis, Visualization, Dashboarding, and Machine Learning Portfolio Project

Portfolio Report

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*Project foundation: IBM Applied Data Science Capstone (SpaceX)
Portfolio reconstruction and methodological extension based on the executed notebooks, archived datasets, dashboard outputs, and analytical artifacts.*

Abstract

This report presents an **end-to-end empirical data analytics project** built around the prediction and interpretation of **SpaceX Falcon 9 first-stage landing outcomes**. The project is organized as an **eight-stage analytical pipeline: historical API acquisition, web scraping, target construction and data wrangling, SQL-based exploratory analysis, visual exploratory data analysis and feature engineering, geospatial analysis, interactive dashboarding, and classification-model evaluation**. The report intentionally synthesizes the analytical logic and empirical findings rather than reproducing notebook cells. Its purpose is to demonstrate how raw, heterogeneous data can be converted into validated decision-relevant evidence through a reproducible **Python-centered workflow**.

Across the 90-mission modeling sample, 60 missions are labeled successful and 30 failure/no-success, yielding a 66.7% historical success rate. **Exploratory analysis** shows that landing outcomes vary substantially by launch site, payload range, orbital category, mission era, and recovery configuration. In the 90-mission sample, KSC LC-39A and VAFB SLC-4E have observed success rates of 77.3% and 76.9%, respectively, compared with 60.0% for CCAFS SLC-40. Payload is associated with success non-linearly: the 8,000+ kg band has the highest observed rate at 87.0%, while the 6,000-8,000 kg band is 50.0%. The annual series shows long-run improvement but not monotonic progress, including a decline from 83.3% in 2017 to 61.1% in 2018 before recovery to 90.0% in 2019 and 84.2% in 2020.

The **predictive stage** substantially extends the comparison. Repeated nested stratified cross-validation indicates that **SVM** has the strongest average accuracy (0.838), F1 (0.884), success recall (0.930), and ROC-AUC (0.878), while the **Decision Tree** has the strongest balanced accuracy (0.800) and failure recall/specificity (0.713). A **time-ordered holdout** of the latest 18 missions reveals why metric choice matters: Logistic Regression, SVM, and Decision Tree each achieve 88.9% accuracy but only 0.667 balanced accuracy because they identify all 15 successes yet only one of three failures. The central methodological **conclusion** is therefore **not that one algorithm is universally superior, but that model ranking depends on the decision objective, class balance, and validation design**.

For a **business analytics** audience, the project demonstrates a **complete analytical value chain**: acquiring data, validating provenance, engineering decision targets, querying structured data, communicating patterns visually and geographically, building interactive interfaces, and evaluating predictive models with attention to generalization and error cost. The strongest portfolio contribution is the **integration** of technical implementation with disciplined interpretation and explicit limits on what the historical data can support.

Keywords

SpaceX; Falcon 9; business analytics; Python; API; web scraping; SQL; exploratory data analysis; feature engineering; Folium; Plotly Dash; classification; nested cross-validation; temporal validation; model evaluation.

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Executive Summary

This portfolio project was developed to demonstrate practical mastery of the **complete data-analysis lifecycle** rather than a single isolated modeling technique. The **analytical question is whether historical mission characteristics can help explain and predict successful Falcon 9 first-stage landing outcomes**. The original IBM capstone provides the project scenario and core workflow; the portfolio version preserves that sequence while strengthening reproducibility, validation, interpretation, and research-style communication. IBM's capstone description explicitly frames the project around data collection, wrangling, exploratory analysis, visualization, model development, model evaluation, and stakeholder reporting [1].

End-to-End Analytical Workflow



Figure 1. End-to-end workflow used in the portfolio project. Each stage corresponds to one of the eight finalized notebooks.

The project deliberately retains multiple historical data snapshots because the workflow uses different prepared datasets for different analytical tasks. The 90-mission sample is the principal modeling and visual-EDA dataset; the SQL stage uses a 101-row IBM-provided table; and the geospatial/dashboard stages use a 56-row historical snapshot with explicit latitude/longitude and dashboard variables. These samples are not silently pooled. Their scope is documented so that findings are interpreted against the correct analytical base.

Stage	Primary artifact	Analytical sample	Role
01 API collection	Historical SpaceX API execution	90 Falcon 9 launches after project filtering	Entity enrichment and reproducible historical snapshot
02 Web scraping	Archived Wikipedia scrape	228 rows	Independent HTML table extraction and validation
03 Wrangling	dataset_part_1.csv	90 rows, 17 source columns	Semantic landing target; 60 successes / 30 failures
04 SQL EDA	Spacex.csv	101 rows, 10 columns	SQLite staging, cleaning, SQL analytical queries
05 Visual EDA	dataset_part_2_portfolio.csv	90 rows, 18 columns	Site/orbit/payload/year analysis; 81-feature matrix
06 Geospatial	spacex_launch_geo.csv	56 rows, 13 columns	Site/outcome maps and proximity analysis
07 Dashboard	spacex_launch_dash.csv	56 rows, 6 analytical columns	Interactive site/payload/booster exploration
08 ML	90-target + 81-feature matrix	90 observations, 81 predictors	Nested CV, temporal holdout, interpretation

Table 1. Stage-specific data snapshots and analytical roles.

1. Motivation and Analytical Framing

1.1 Business problem

The capstone scenario treats **first-stage recovery** as economically consequential because reusable launch hardware changes the cost structure of orbital launch services. From a business analytics perspective, the **central problem** is therefore not merely to classify a binary outcome. It is to convert operational history into evidence that could support bidding, risk assessment, resource planning, and competitive decision-making. The classification target functions as an analytically tractable proxy for a broader managerial question: **under what mission configurations and operating regimes has successful recovery historically been more likely?**

The report adopts a **decision-oriented interpretation**. A false prediction of success and a false prediction of failure may have different business consequences. Consequently, the machine-learning stage does not rank models by accuracy alone. It also evaluates balanced accuracy, success recall, failure recall/specificity, precision, F1, and ROC-AUC, and it examines whether model behavior persists when the validation scheme respects time order.

1.2 Research questions

- How can heterogeneous launch data be acquired, validated, and transformed into a reproducible analytical dataset?
- Which mission, site, orbit, payload, temporal, and recovery characteristics are associated with observed landing success?
- How can SQL, statistical graphics, geographic visualization, and interactive dashboards provide complementary views of the same operational problem?
- Do Logistic Regression, SVM, Decision Tree, and KNN differ meaningfully once evaluation moves beyond a single small train/test split?
- How does temporal non-stationarity affect the interpretation of predictive accuracy and failure detection?

1.3 Report architecture

The report follows a research-report structure of problem framing, methodology, results, discussion, limitations, and conclusion. This organization is consistent with university guidance that research reports normally separate introduction, methodology, results, discussion, and conclusions [2], as well as MIT guidance describing the common scientific sequence of title/abstract/introduction/methods/results/discussion/references [3]. External sources are used here only to guide report structure and methodological framing; all empirical SpaceX results in the report come from the finalized project notebooks and associated CSV/output artifacts.

2. Data Acquisition and Provenance

2.1 Historical SpaceX API acquisition

Notebook 01 uses the **SpaceX REST API** as the first acquisition channel. The launch endpoint supplies the base launch records, and entity identifiers are resolved through related endpoints to recover rocket, payload, launchpad, and core information. In the original successful execution, identifiers were translated into analytically useful variables such as booster version, payload mass and orbit, launch-site coordinates, block, reuse count, serial number, and landing-related attributes. After the Falcon 9 project filter, the historical working dataset contains 90 launches.

A **reproducibility constraint** emerged after the original project was completed: the API endpoints used in the historical run later became unavailable (June 2026). Rather than replacing the original outputs with unrelated contemporary data, the portfolio preserves the successfully executed API notebook and archived API response. This is methodologically important because reproducibility is not synonymous with live re-queryability; for historical analyses, a fixed source snapshot can be the more defensible basis for downstream replication when an external service disappears.

2.2 Web scraping as an independent acquisition method

Notebook 02 demonstrates a second acquisition pathway using **HTML table scraping**. The final portfolio notebook identifies candidate tables by schema, uses reusable parsing functions, validates shape and identifiers, and retains an archived mode so the result can be reproduced without depending on a mutable web page.

The validated archived scrape contains 228 rows and 11 columns, with no duplicate rows, flight numbers ranging from 1 to 121, five launch-site labels, and eight orbit labels. The role of this stage is methodological as much as substantive: it demonstrates the ability to turn semi-structured web content into a checked tabular dataset and to distinguish extraction logic from later analytical modeling.

3. Data Wrangling and Target Construction

3.1 Schema and quality controls

Notebook 03 loads the 90-row, 17-column analytical source and performs explicit checks for required columns, unexpected columns, duplicate missions, data types, date conversion, and missingness. It also profiles launch-site and orbit frequencies before constructing the supervised-learning target. These checks are important because the downstream model is only as credible as the semantic definition of the response variable.

3.2 Semantic construction of landing success

A key portfolio improvement over IBM capstone concerns target engineering. The original capstone logic identified unsuccessful outcomes using positional indices from a frequency table, which is fragile because frequency order can change. The revised notebook defines the binary target semantically: landing outcomes beginning with 'True' are mapped to Class = 1, while outcomes beginning with 'False' or 'None' are mapped to Class = 0. Unexpected or missing target outcomes raise an error. This converts an implicit, order-dependent rule into an explicit and testable business definition.

The resulting dataset contains 90 missions and 18 columns. There are 60 successful landings and 30 failure/no-success outcomes, so the historical success rate is 66.67%. This imbalance is moderate but material: a majority classifier that always predicts success already obtains 66.7% accuracy, which later motivates the use of balanced accuracy and class-specific recall.

4. SQL-Based Exploratory Analysis

4.1 Relational workflow

Notebook 04 intentionally retains the SQL-specific IBM dataset. The 101-row, 10-column `'Spacex.csv'` file is loaded into pandas, written to a SQLite staging table (`'SPACEXTBL'`), and transformed into a **cleaned** analysis table (`'SPACEXTABLE'`) with non-null dates.

4.2 SQL findings

SQL question / measure	Result
Launch-site coverage	Four labels: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
NASA CRS payload	45,596 kg total payload
F9 v1.1 payload	2,534.67 kg average payload
First successful ground landing	2015-12-22
Successful drone-ship landings, 4,000–6,000 kg	F9 FT B1021.2, B1022, B1026, B1031.2
Mission outcome counts	100 successful missions; 1 mission failure
Maximum payload	15,600 kg
2015 failed drone-ship landings	B1012 (Jan 10) and B1015 (Apr 14), both CCAFS LC-40

Table 2. Selected SQL results produced from the finalized Notebook 04.

The SQL stage illustrates an important distinction between mission outcome and first-stage landing outcome. The 100-versus-1 mission-success result does not mean that 100 first stages landed successfully. Mission success concerns delivery of the primary mission objective, whereas the recovery target concerns the first-stage landing result. Preserving this distinction prevents a common analytical category error.

5. Exploratory Data Analysis and Feature Engineering

5.1 Launch-site effects

In the 90-mission modeling sample, **launch-site performance** differs substantially. KSC LC-39A records 17 successes in 22 missions (77.3%), VAFB SLC-4E records 10 in 13 (76.9%), and CCAFS SLC-40 records 33 in 55 (60.0%). These are descriptive rates, not causal estimates: site choice is intertwined with mission era, payload, orbit, booster generation, and recovery strategy.

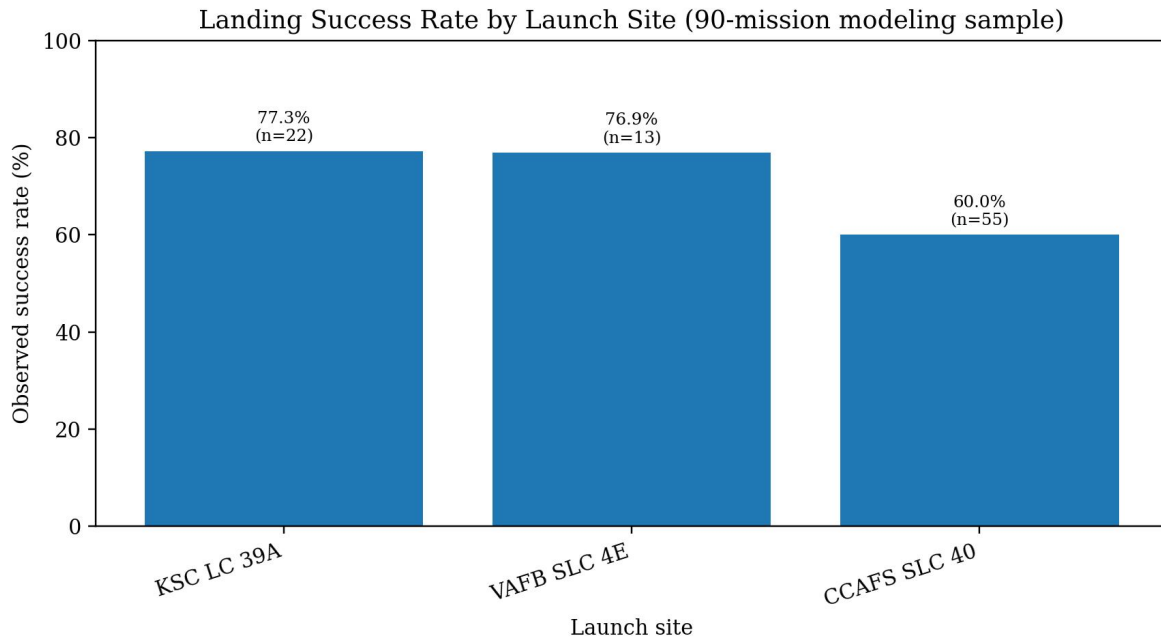


Figure 2. Observed first-stage landing success rate by launch site in the 90-mission modeling sample.

5.2 Payload mass is associated with success non-linearly

Payload mass shows a meaningful but non-monotonic association with landing success. The 0-2,000 kg band has a 58.3% observed success rate (7/12), the 2,000-4,000 kg band 69.2% (18/26), the 4,000-6,000 kg band 52.9% (9/17), and the 6,000-8,000 kg band 50.0% (6/12). The 8,000+ kg group is highest at 87.0% (20/23). The appropriate interpretation is therefore not that heavier payloads uniformly increase or decrease success; payload effects must be read jointly with other mission characteristics and historical program maturity.

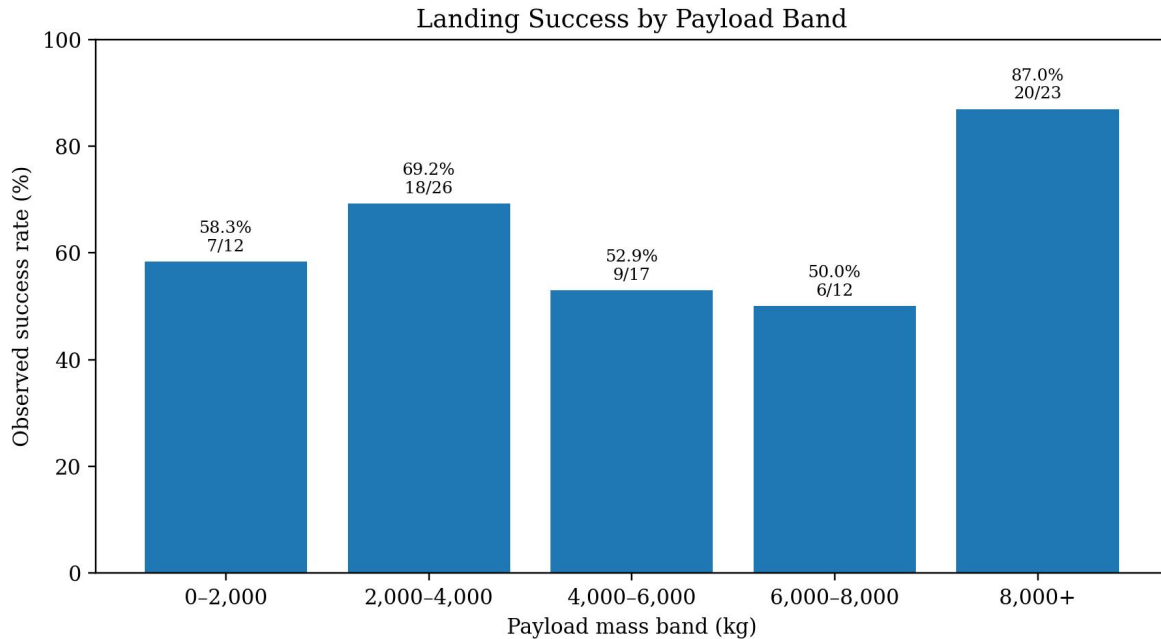


Figure 3. Landing success across payload-mass bands. The heaviest band has the highest observed rate, demonstrating the non-linear pattern.

5.3 Temporal learning and non-stationarity

Landing success improves markedly over the decade, but not monotonically. The annual success rate is 0% in 2010, 2012, and 2013; 33.3% in both 2014 and 2015; 62.5% in 2016; 83.3% in 2017; 61.1% in 2018; 90.0% in 2019; and 84.2% in 2020. This pattern is consistent with a maturing recovery program while also showing year-to-year variation. The temporal structure becomes central to the model-evaluation design in Notebook 08.

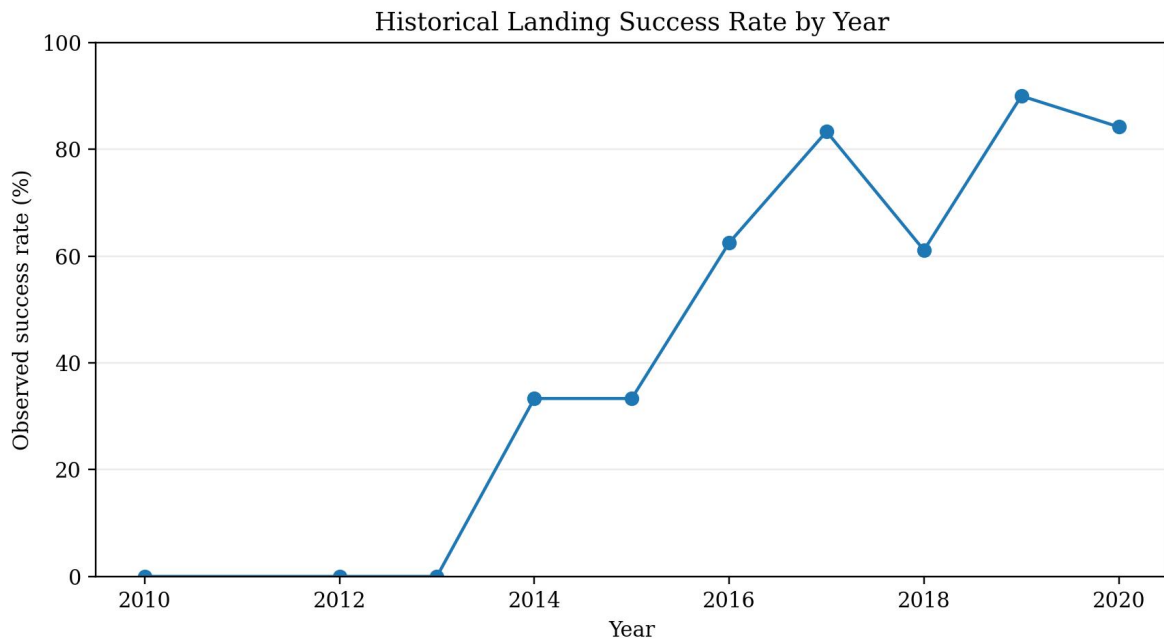


Figure 4. Historical success rate by year in the modeling sample. Improvement is strong in the long run but not monotonic.

5.4 Feature engineering

Categorical variables are **one-hot encoded** after explicitly representing missing landing-pad information as 'No landing pad'. The finalized feature matrix contains 90 rows and 81 fully numeric columns, with no missing values and no duplicate feature rows. This matrix includes continuous/ordinal operational variables such as FlightNumber, PayloadMass, Flights, Block, and ReusedCount; binary configuration indicators such as GridFins, Reused, and Legs; and encoded orbit, launch-site, landing-pad, and core-serial categories. The matrix is passed to Notebook 08 without altering row order.

6. Geospatial Analysis

6.1 Purpose and sample

Notebook 06 uses the 56-row geospatial snapshot because it contains the coordinates required for **Folium** analysis. The sample covers four launch-site labels and produces **site-level landing summaries** consistent with the dashboard dataset: KSC LC-39A 76.9% (10/13), CCAFS SLC-40 42.9% (3/7), VAFB SLC-4E 40.0% (4/10), and CCAFS LC-40 26.9% (7/26). The difference from Notebook 05's site rates is expected because the analytical snapshots are different.

6.2 Interactive mapping and proximity reasoning

The **Folium** workflow maps **launch sites** and **clustered success/failure outcomes**, adds cursor coordinates, and uses great-circle (haversine) distance calculations for a CCAFS LC-40 proximity case study. The final notebook uses an Esri World Street Map background for readable city and road context.

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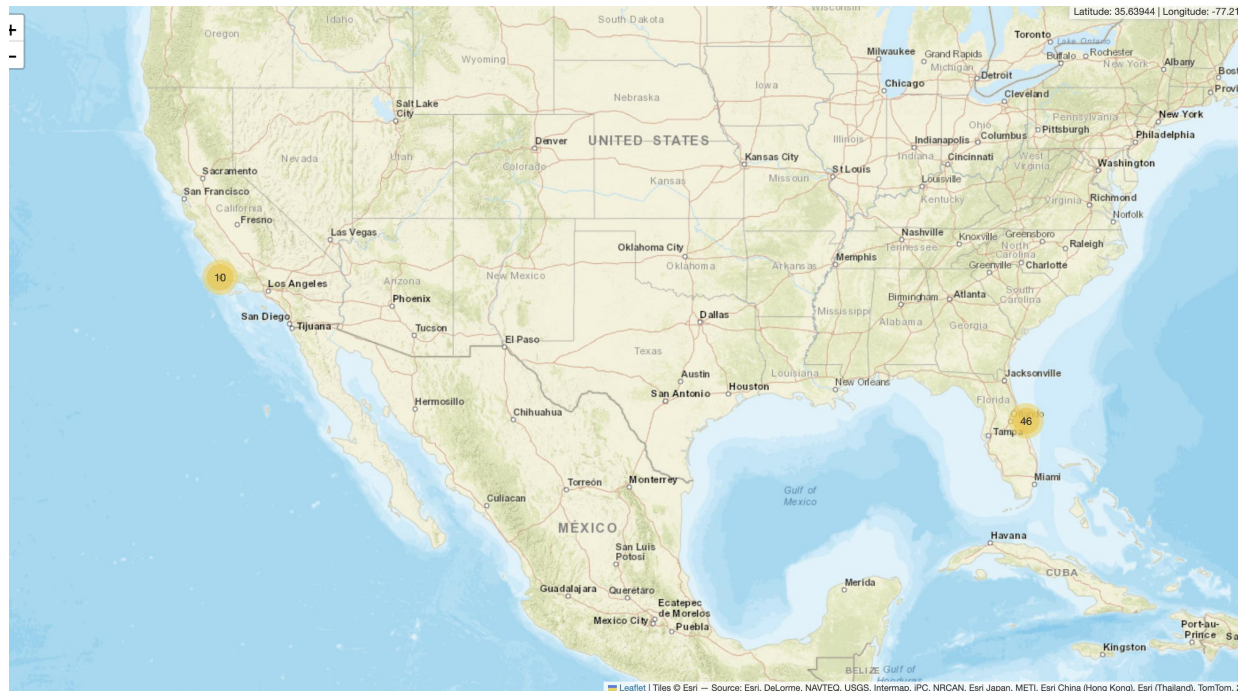


Figure 5. Folium clustered launch sites map screenshot from the project output. Number represents the count of launch sites near that area.

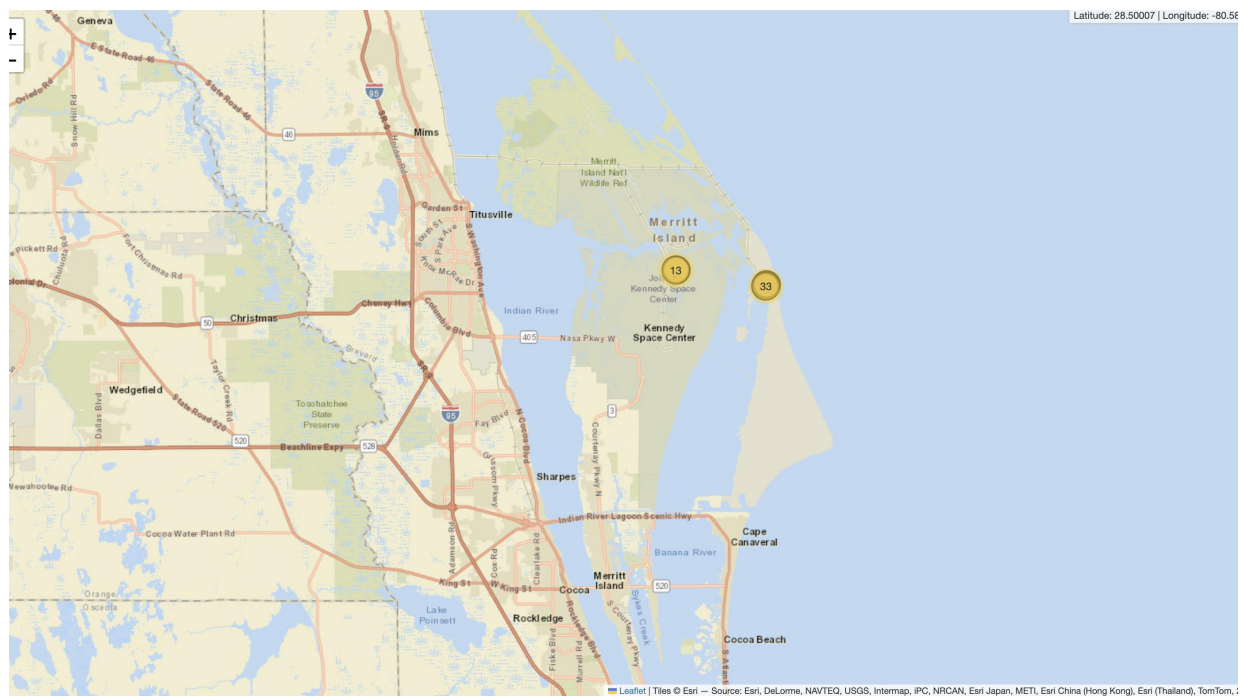


Figure 6. Folium clustered launch sites map screenshot from the project output. A zoomed in display of the Kennedy Space Center.

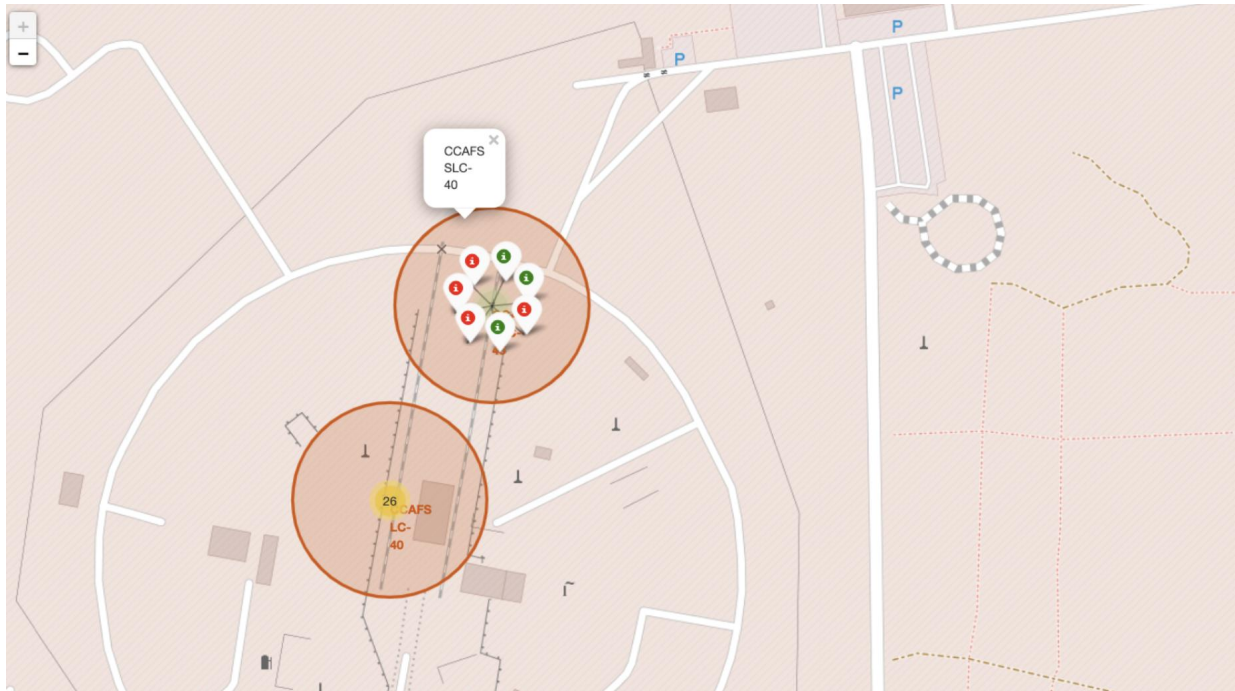


Figure 7. Folium launch-outcome map screenshot from the project output. Red/green markers represent historical failure/success outcomes; the finalized Notebook 06 preserves the analytical markers while using an Esri street basemap.

The proximity case study uses selected reference coordinates. The selected **coastline** reference point is approximately 0.58 km from CCAFS LC-40, and the selected nearby **city** point is approximately 23.16 km away. The result indicates that launch site show close proximity near coastline and keeps certain distance from the city..

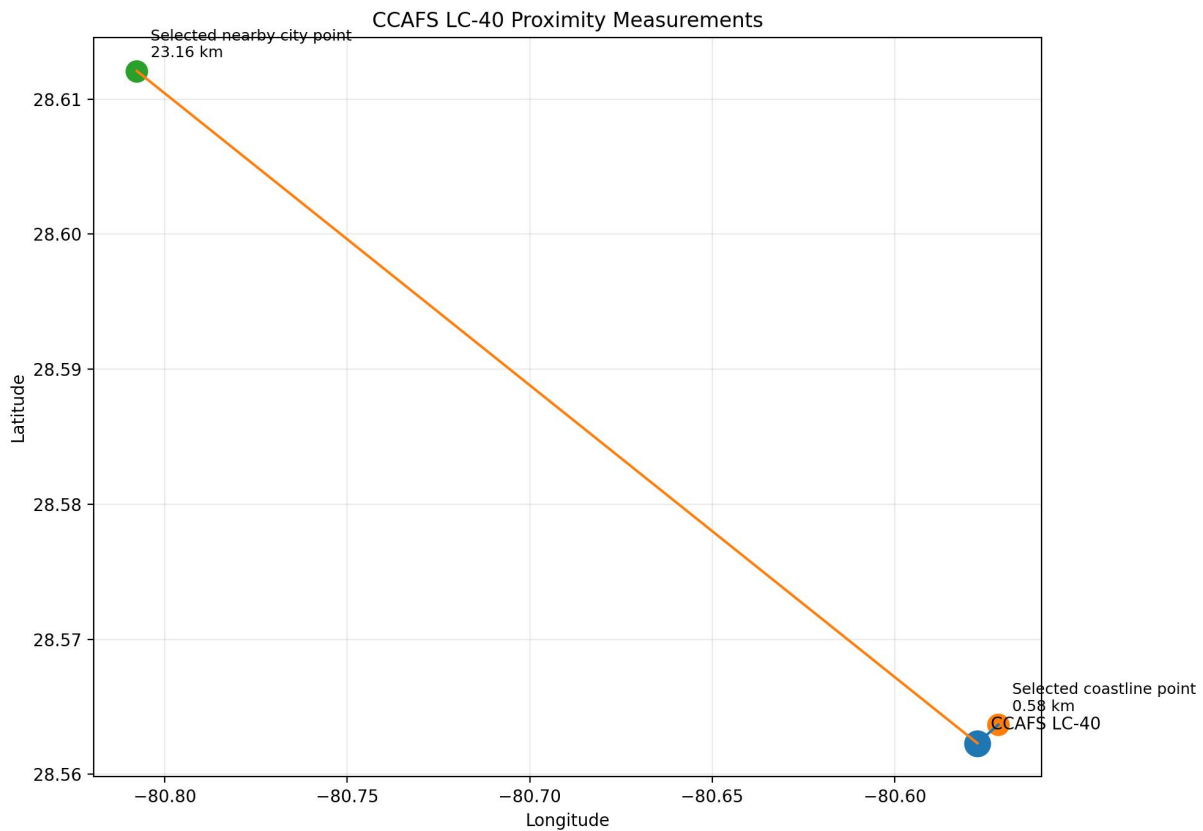


Figure 8. CCAFS LC-40 proximity measurements: selected coastline reference approximately 0.58 km; selected nearby city reference approximately 23.16 km.

7. Interactive Dashboard and Decision Support

7.1 Dashboard design

Notebook 07 turns the 56-row dashboard snapshot into an **interactive Plotly Dash interface**. The user can select a launch site, restrict the payload range, view a success/outcome pie chart, and inspect payload mass against landing class with booster-version category as an additional encoding. The portfolio separates reusable figure logic from callback wiring, generates site options from the data, adds empty-filter handling, and exports a standalone Python(.py) application.

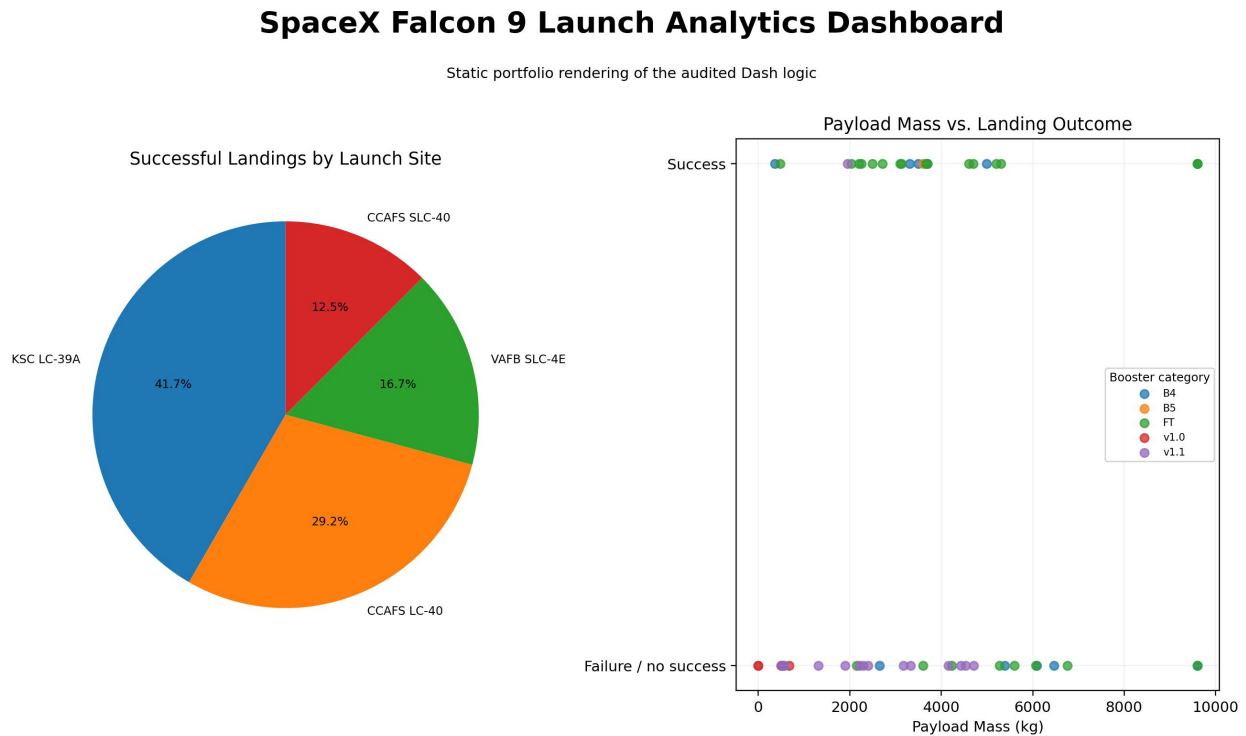


Figure 9. Static rendering of the audited Dashboard 07 analytical logic: successful landings by launch site and payload-versus-outcome scatter by booster category.

7.2 Dashboard findings

Dashboard question	Audited answer
Largest number of successful landings	KSC LC-39A: 10
Highest observed site success rate	KSC LC-39A: 76.9% (10/13)
Highest fixed 2,000 kg payload-band rate	2,000–4,000 kg: 61.9% (13/21)
Lowest fixed payload-band rate	6,000–8,000 kg: 0% (0/4)
Highest nominal booster-category rate	B5: 100% (1/1)
Highest multi-observation booster-category rate	FT: 66.7% (16/24)

Table 3. Audited dashboard findings from the 56-mission snapshot.

The B5 result illustrates why dashboard interaction must be accompanied by statistical judgment. A 100% rate based on one mission is not comparable in evidential strength to a 66.7% rate based on 24 FT missions.

The dashboard is therefore best viewed as an exploratory decision-support interface rather than a causal or inferential model.

8. Machine Learning and Model Evaluation

8.1 Why the original 83.33% tie required deeper evaluation

The original capstone notebook tuned **Logistic Regression**, **SVM**, **Decision Tree**, and **KNN** and evaluated them on one 18-observation test set. All four obtained **83.33% test accuracy**. A tie on such a small holdout does not establish that the models are equivalent; it can conceal different error structures and substantial sampling variability. The **portfolio** notebook therefore treats the original split as a benchmark, not as the primary selection mechanism.

Instead, hyperparameters are selected by **balanced accuracy**, which scikit-learn defines as the average recall across classes and recommends for imbalanced classification settings [4].

8.2 Baseline and evaluation metrics

Because 60 of 90 missions are successes, a majority-class **baseline** that always predicts success produces 66.7% accuracy, 0.50 balanced accuracy, and 0.50 ROC-AUC. This baseline exposes a central risk: accuracy can look respectable even when a model has no ability to distinguish failures. The report therefore emphasizes balanced accuracy and failure recall/specificity in addition to success-oriented metrics.

8.3 Repeated nested stratified cross-validation

The **primary comparison** uses a **5-fold stratified outer loop repeated five times**, yielding 25 out-of-sample evaluations per model. A four-fold stratified inner loop selects hyperparameters using **balanced accuracy**. This design separates hyperparameter selection from outer-loop performance estimation. Scikit-learn's methodological guidance notes that non-nested tuning and evaluation can produce optimistically biased scores, whereas nested cross-validation estimates the generalization performance of both the fitted model and its hyperparameter search [5].

Model	Accuracy	Balanced acc.	Failure recall	Success recall	F1	ROC-AUC
Decision Tree	0.829	0.800	0.713	0.887	0.873	0.846
SVM	0.838	0.792	0.653	0.930	0.884	0.878
Logistic Regression	0.807	0.767	0.647	0.887	0.858	0.864
KNN	0.800	0.737	0.547	0.927	0.861	0.834

Table 4. Mean repeated nested cross-validation performance (25 outer evaluations per model).

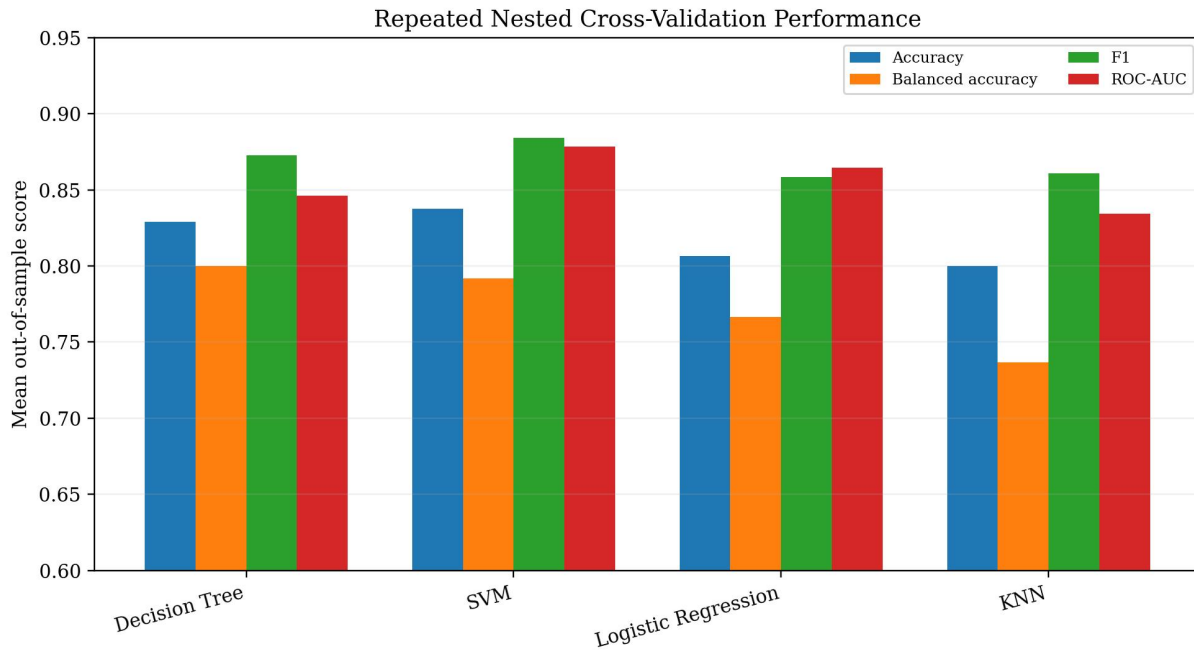


Figure 10. Repeated nested cross-validation performance. SVM is strongest on average accuracy/F1/ROC-AUC, while Decision Tree leads balanced accuracy.

The models are similar but not identical. **SVM** has the **highest mean accuracy (0.838)**, F1 (0.884), success recall (0.930), and ROC-AUC (0.878). **Decision Tree** has the **highest mean balanced accuracy (0.800)** and failure recall/specificity (0.713). **Logistic Regression** remains competitive with 0.864 ROC-AUC, while **KNN** has the **weakest** mean balanced accuracy (0.737) and failure recall (0.547). The standard deviation of balanced accuracy across outer evaluations ranges from approximately 0.075 to 0.090, so the evidence does not justify claiming a decisive universal winner.

8.4 Temporal holdout

The historical success trend suggests non-stationarity, so the final evaluation sorts missions by **launch date**, trains on the earliest 72 missions, and reserves the latest 18 missions—from 2020-01-19 through 2020-11-05—as an untouched temporal holdout. Hyperparameters are selected within the earlier period using an expanding **TimeSeriesSplit**. The holdout contains 15 successes and only three failures, meaning that an always-success baseline already achieves 83.33% accuracy.

Model	Accuracy	Balanced acc.	Failure recall	Success recall	F1	ROC-AUC
Logistic Regression	0.889	0.667	0.333	1.000	0.938	0.589
SVM	0.889	0.667	0.333	1.000	0.938	0.533
Decision Tree	0.889	0.667	0.333	1.000	0.938	0.667
KNN	0.833	0.500	0.000	1.000	0.909	0.589
Majority Baseline	0.833	0.500	0.000	1.000	0.909	0.500

Table 5. Temporal holdout results for the latest 18 missions.

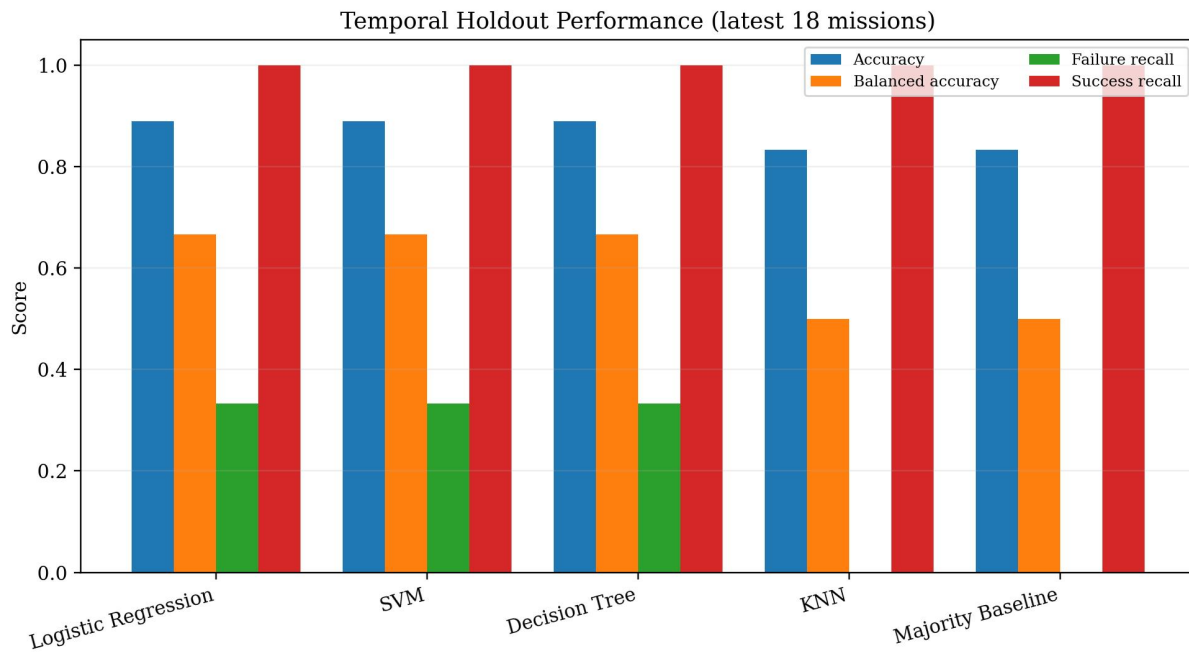


Figure 11. Temporal holdout metrics. Overall accuracy remains high because the latest sample is success-dominated; failure recall exposes the remaining weakness.

Logistic Regression, SVM, and Decision Tree each obtain **88.89% accuracy** and 0.667 balanced accuracy. All three correctly classify all 15 later successes but identify only one of the three failures; two failures are incorrectly predicted as successes. KNN predicts every later mission as a success and therefore matches the majority baseline in class discrimination: 83.33% accuracy, 0.50 balanced accuracy, and zero failure recall. This result is more important than the increase from 83.3% to 88.9% raw accuracy: the temporal experiment shows that high headline accuracy can coexist with weak recognition of the operationally rarer failure class.

8.5 Interpretable predictive signals

The **post-evaluation interpretation** stage fits tuned Logistic Regression and Decision Tree models on the full historical sample solely to examine **predictive associations**. Among the strongest standardized **logistic coefficients** are 'Legs' (+), 'LandingPad_No landing pad' (-), 'GridFins' (+), 'ReusedCount' (+), 'Block' (+), and 'FlightNumber' (+). The tuned shallow **Decision Tree** concentrates most importance on 'Legs', 'ReusedCount', 'Reused', 'PayloadMass', and 'Orbit_ISS'.

These variables should not be read as causal drivers. Several encode recovery configuration or program maturity and may partly capture whether a landing was planned, whether the vehicle belongs to a later operating generation, or which mission regime is being observed. Their value here is predictive and descriptive. A causal study would require a different research design and stronger assumptions about confounding and treatment assignment.

9. Integrated Discussion

9.1 What the project demonstrates analytically

The strongest feature of the portfolio is not any single algorithm. It is the **continuity of reasoning from acquisition to evaluation**. The project starts with raw API entities and semi-structured HTML, converts them into checked tables, constructs an explicit business target, queries structured data with SQL, explores

patterns visually and geographically, exposes the analysis through an interactive interface, engineers a model matrix, and finally evaluates predictive models under multiple validation designs. This is the type of **end-to-end chain** that **business analytics research** often requires: decisions depend not only on a final model but also on data provenance, variable definitions, diagnostics, and the communication layer through which results reach stakeholders.

9.2 Descriptive evidence and predictive evidence are complementary

EDA indicates that landing success varies by site, payload band, orbit, booster/recovery configuration, and time. Machine learning combines those signals but does not make the descriptive findings obsolete. In fact, the temporal trend explains why the temporal model results must be interpreted cautiously: later missions are much more likely to succeed, so a classifier can obtain high accuracy by learning the mature operating regime while still being poor at distinguishing rare failures.

This reinforces a broader business analytics principle: **model metrics should be read in the context of the underlying process**. The apparent 88.9% temporal accuracy is less decision-useful than the fact that two out of three failures were missed by three of the four substantive models. If false-success predictions carry greater financial or operational cost than false-failure predictions, model selection, class weights, thresholds, or loss functions should be adjusted accordingly.

9.3 Business relevance

The project translates naturally into several **managerial use cases**. First, a launch-services bidder could use recovery likelihood as one input to **expected-cost and risk scenarios**. Second, **operations analysts** could use site/payload/booster dashboards to identify mission configurations that warrant deeper review. Third, **model monitoring** could focus not only on accuracy but on failure recall and calibration as the operating environment changes. Finally, the pipeline demonstrates **governance practices**—historical snapshots, auditable target rules, stage-specific datasets, and explicit validation—that are transferable to financial analytics, supply-chain risk, customer churn, credit decisions, and other business problems where data drift and asymmetric error costs matter.

10. Limitations and Research Extensions

10.1 Limitations

- Small modeling sample: 90 missions limit statistical precision and make high-dimensional one-hot encoding vulnerable to instability.
- Class imbalance: success is 66.7% of the full modeling sample and 83.3% of the temporal holdout.
- Temporal non-stationarity: the recovery program matured strongly over time, so random folds can mix operating regimes that would be separated in real forecasting.
- Stage-specific snapshots: SQL, geospatial, and dashboard stages use historical datasets of different sizes. The report treats these as separate analytical snapshots rather than assuming direct row-wise equivalence.
- Feature timing and meaning: variables such as landing configuration are highly predictive, but a production model must specify exactly what information is available at the moment a prediction is made.
- Observational design: feature coefficients and tree importances are associations, not causal effects.
- External-source reproducibility: the original API later became unavailable; reproducibility therefore relies on preserved historical outputs and archived project data.

10.2 Extensions appropriate for further research

A larger longitudinal dataset would allow more stable temporal validation, calibration analysis, probabilistic scoring, and subgroup error analysis. A cost-sensitive extension could assign explicit economic penalties to false-success and false-failure predictions and tune the classification threshold accordingly. Survival or event-history perspectives could model technological learning across sequential booster reuse. Hierarchical models could partially pool information across sites, orbits, and booster generations. Finally, a causal design could ask a different question—such as the effect of specific recovery technologies—provided that treatment timing, confounding, and selection mechanisms were explicitly modeled.

11. Reproducibility and Portfolio Governance

The finalized portfolio uses numbered notebooks as the authoritative analysis sequence: **01 API Data Collection**, **02 Web Scraping**, **03 Data Wrangling**, **04 SQL/SQLite EDA**, **05 Visual EDA and Feature Engineering**, **06 Folium Geospatial Analysis**, **07 Plotly Dash Dashboard**, and **08 Machine Learning Prediction/Classification**. Supporting CSV, JSON, SQLite, HTML, PNG, and Python files are retained with the stage that generated or consumed them.

Several deliberate safeguards improve **reproducibility**: archived API outputs are preserved when the live service is unavailable; web scraping has an archived mode; target construction uses semantic labels rather than frequency positions; and Notebook 08 exports a CSV of exact model-performance results for direct use in reporting. The report itself is based on those finalized artifacts, not on substituted external SpaceX facts.

12. Conclusion

This project demonstrates a **complete Python-centered analytical workflow from raw acquisition to model evaluation**. The technical contribution is breadth with continuity: REST API work and web scraping feed into validation and wrangling; SQL provides a relational analytical layer; EDA and feature engineering reveal structure and prepare the model matrix; geospatial and dashboard components expand how findings are communicated; and the machine-learning stage evaluates multiple classifiers using a validation framework appropriate to a small, imbalanced, temporally evolving dataset.

Empirically, the historical record shows clear **improvement in landing success over time**, **substantial variation across mission contexts**, and a **non-linear relationship with payload mass**. Predictively, the strongest conclusion is conditional rather than absolute. **SVM** provides the strongest average ranking/discrimination performance in repeated nested validation, while the **Decision Tree** performs best on balanced accuracy and failure recall. Temporal testing exposes a more fundamental challenge: **the later operating period is dominated by success, so high overall accuracy can mask poor identification of the rarer failures**.

For a portfolio in **business and data analytics**, the project is therefore evidence of more than software proficiency. It shows the ability to formulate a decision problem, manage data provenance, diagnose and correct analytical weaknesses, reconcile results across multiple representations, evaluate models beyond headline accuracy, and communicate conclusions with appropriate limits. Those are core research capabilities for empirical business analytics.

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Appendix A. Finalized Analytical Artifacts

Stage	Final notebook	Primary contribution
01	01_SpaceX_API_Data_Collection_Portfolio.ipynb	Preserved historical API code and outputs
02	02_SpaceX_Web_Scraping_Portfolio.ipynb	Validated archived/live scraping workflow
03	03_SpaceX_Data_Wrangling_Portfolio.ipynb	Semantic target construction; dataset_part_2_portfolio.csv
04	04_SpaceX_EDA_SQL_Portfolio.ipynb	SQLite staging and SQL EDA
05	05_SpaceX_EDA_Data_Visualization_Portfolio.ipynb	Audited EDA; dataset_part3_portfolio.csv
06	06_SpaceX_Folium_Launch_Site_Analysis_Portfolio.ipynb	Esri-basemap geospatial analysis and HTML maps
07	07_SpaceX_Dash_Interactive_Dashboard_Portfolio.ipynb	Dash analytics and standalone app
08	08_SpaceX_Machine_Learning_Portfolio.ipynb	Nested CV, temporal holdout, interpretability

Table A1. Authoritative notebook sequence used to prepare this report.

Appendix B. Key Numerical Results for Reuse

B.1 Modeling-sample site success

Launch site	Launches	Successes	Success rate
KSC LC 39A	22	17	77.3%
VAFB SLC 4E	13	10	76.9%
CCAFS SLC 40	55	33	60.0%

B.2 Modeling-sample payload bands

Payload band (kg)	Launches	Successes	Success rate
0–2,000	12	7	58.3%
2,000–4,000	26	18	69.2%
4,000–6,000	17	9	52.9%
6,000–8,000	12	6	50.0%
8,000+	23	20	87.0%

B.3 Annual landing success

Year	Launches	Successes	Success rate
2010	1	0	0.0%
2012	1	0	0.0%
2013	3	0	0.0%
2014	6	2	33.3%
2015	6	2	33.3%
2016	8	5	62.5%
2017	18	15	83.3%
2018	18	11	61.1%
2019	10	9	90.0%
2020	19	16	84.2%